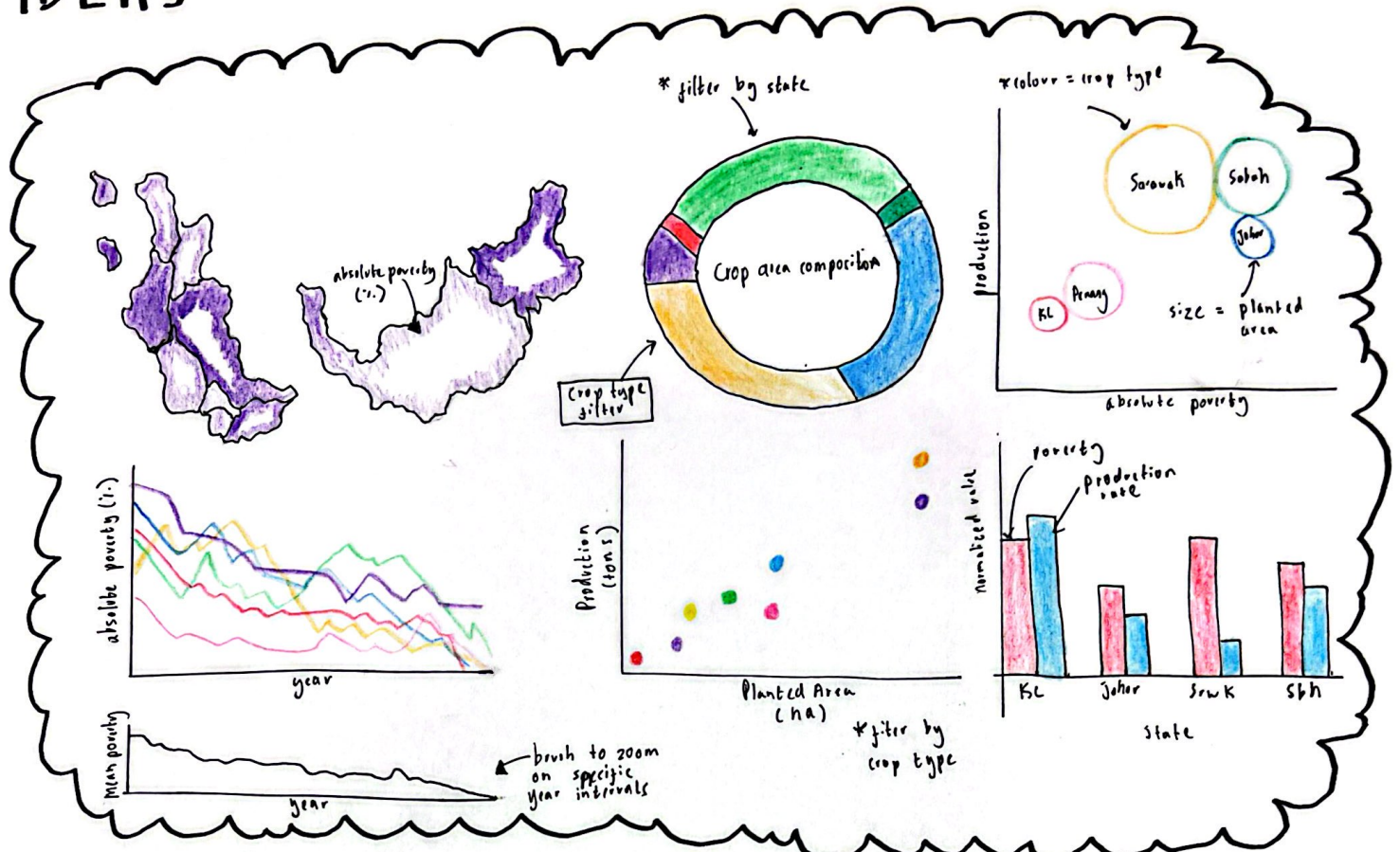




FILTER

- Option to combine choropleth map with symbols to encode poverty & planted area simultaneously
- Donut chart to add filter to display crop area composition by year & state
- Grouped bar chart can become dual axis → removing the need to normalize
- Bubble chart allows comparison for crop type, production, planted area & poverty simultaneously

CATEGORISE

- Spatial** → Choropleth map, can cross cut w/ agriculture data, bubble chart
- Temporal** → Slidable time series graph, can include crop data too
- Composition** → Donut chart; can make two for each poverty & agriculture
- Distribution** → Scatter plot, grouped bar chart; can make scatter y-axis poverty to.

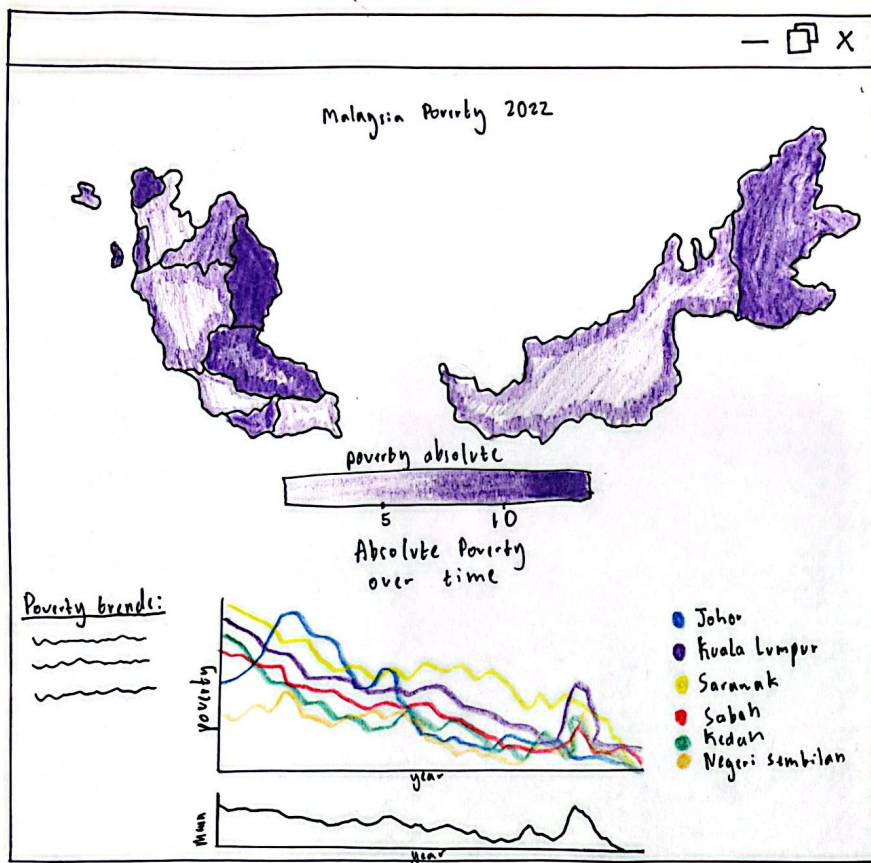
COMBINE & REFINE

- Dashboard 1: Poverty story**
 - Choropleth map (absolute poverty states)
 - Poverty time series per state (change with time)
- Dashboard 2: Agricultural trends**
 - Symbol map (show dominating crop)
 - Donut chart (crop composition per state)
 - Scatter plot (showing production & ha)
- Dashboard 3: Poverty + Agriculture**
 - Bubble chart to show multiple values & groupings
 - Grouped bar chart to explore agricultural outcome & poverty

SUMMARIZE & QUESTION

Questions: What are the relations (if any) with poverty & agricultural output / planted area within states in Malaysia

- Do more impoverished states have higher or lower agricultural activity, does it differ with type of crop
- How does poverty evolve over time across states with different agricultural compositions
- Which type of crops dominate Malaysia's agricultural landscape
- Which crop types dominate in higher poverty states, do



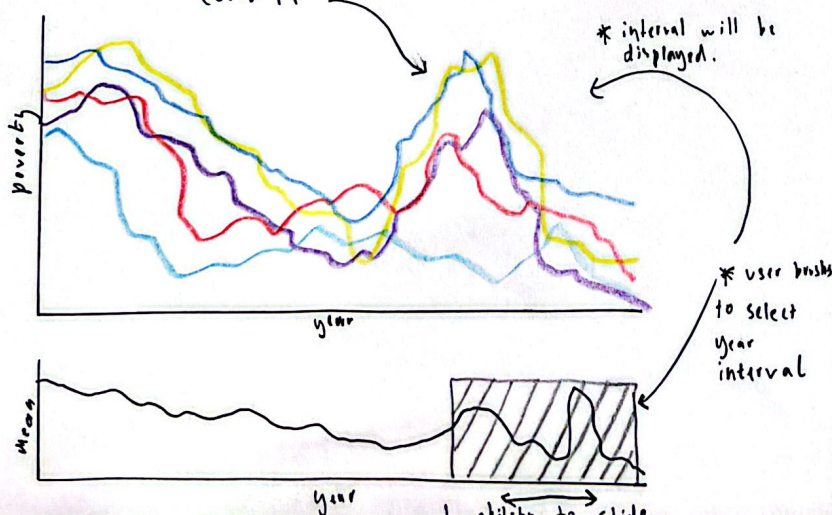
focus



* User can toggle year, poverty type

* Data collection for states does not start consistently

* Annotation & Covid 19



Jamie Iwanie Rajan

Title : Poverty pattern within states in Malaysia

Description: Explore the landscape of poverty in Malaysia

components

Components

- o Choropleth Map of Malaysia
- o Interactive time series chart
- o Several toggles for poverty type

Operations

- o Interactive time series chart includes brushing year interval
- o toggle by poverty type

Details, pros-cons

Pros: Combines spatial & temporal perspectives, allowing viewers to explore where poverty is highest & how it changes over time

Cons: Cognitive overload if not well-coordinated, where users may struggle to see patterns.

Jamie Ivanice Rajem

Title: Crop Area & Production by state

Description: Exploring crop area and it's production for Malaysia and it's states.

Components & Operations

Components

- Donut chart showing crop composition for a particular state
- Scatter plot showing all states for a particular crop

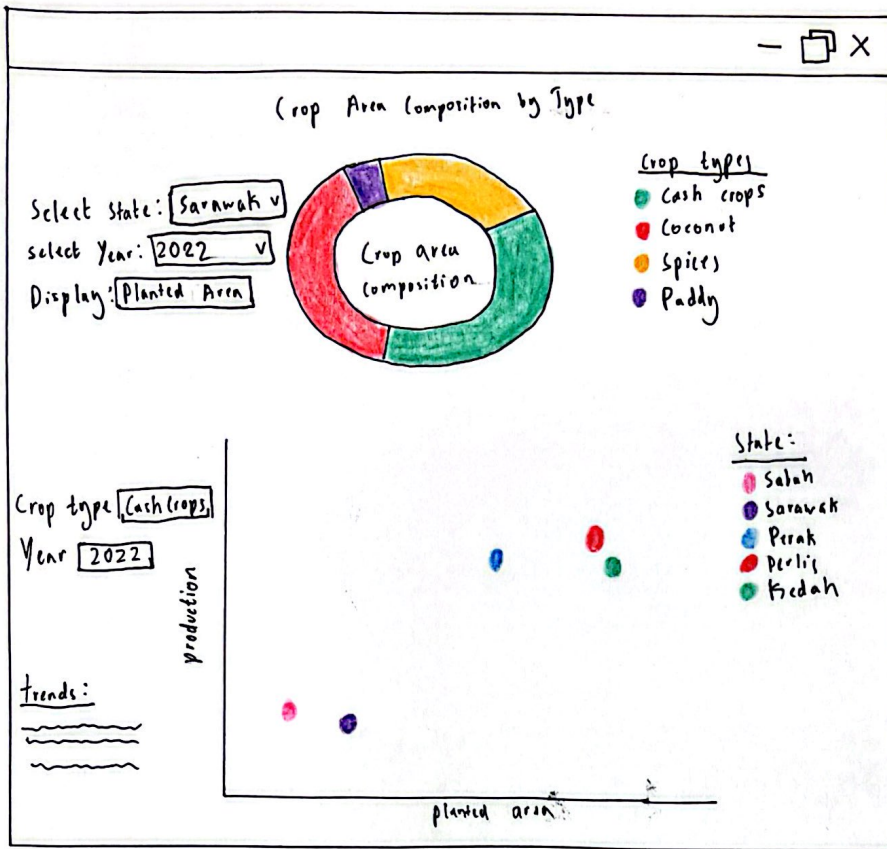
Operations:

- ability to filter donut chart by state & year
- ability to filter scatter plot by crop type & year
- hover options.

Details, Pros & Cons

Pros: Provides composition & state performance, providing good comparative & analytical exploration.

Cons: Different data granularities (aggregate vs per-crop) can confuse viewers, making it hard to relate



focus

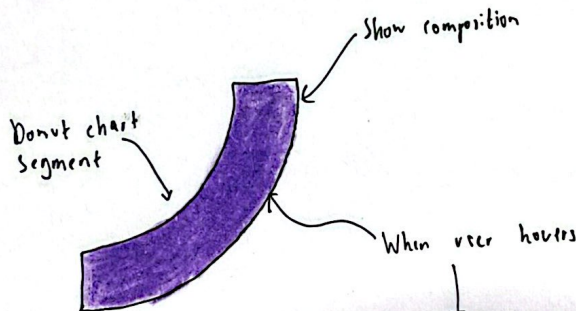
* Display switcher

Display: Production v
Planted Area

* Other filters include year & state

Same filters but now can filter by crop type

Cash Crops

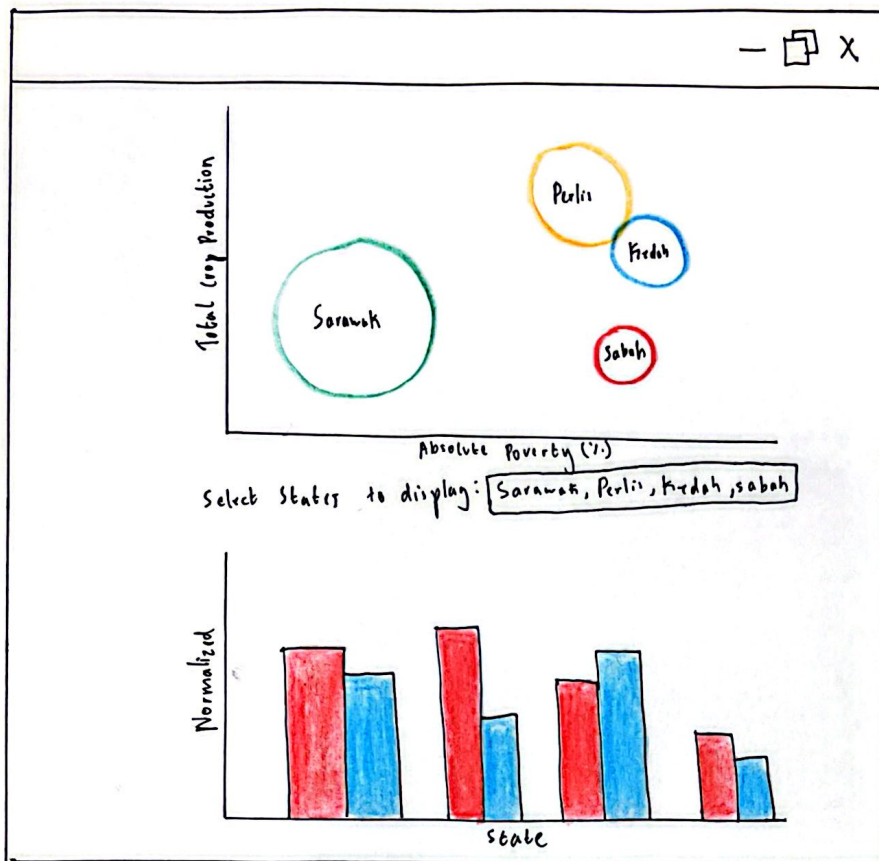


* Scatter plot

* Current state in Malaysia
* for a certain crop

Upon user hover

Selangor, Cash crops
o Planted Area: 9696.9
o Production: 40,000



Jamie Ivorian Rojem

Title: Agricultural Output & Poverty Patterns across Malaysian States

Description: Comparative insight through exploring Malaysia's poverty & Agricultural trends.

Components & Operations

Components

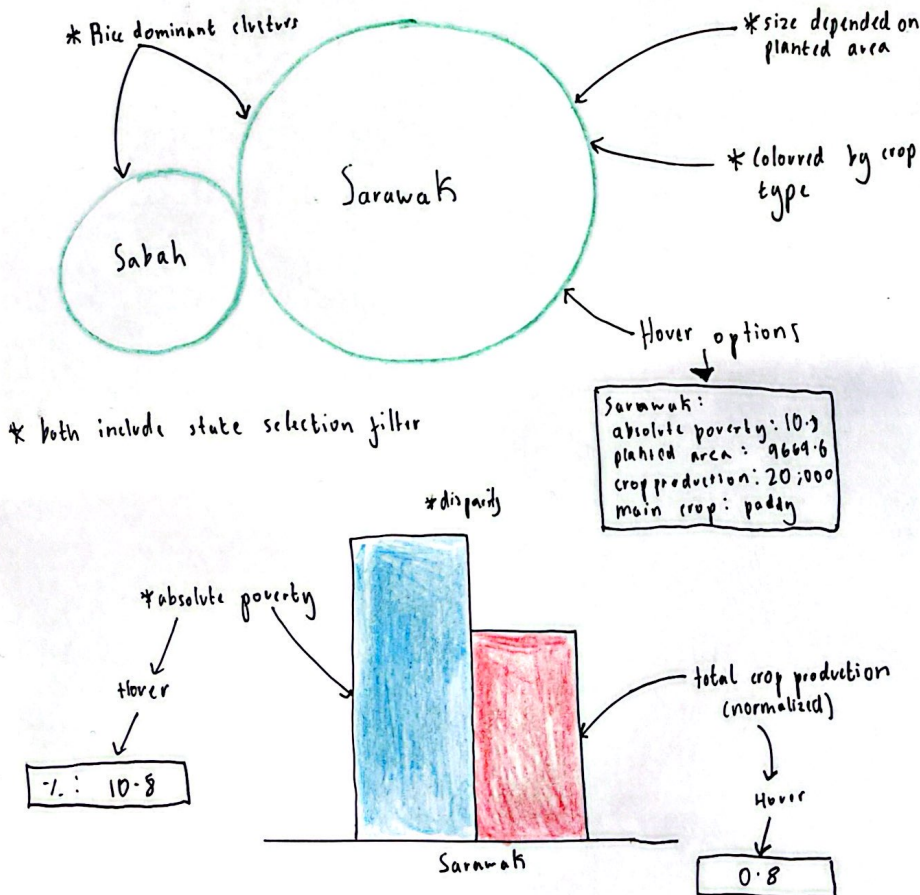
- o Bubble chart to show state clusters, and visually display dominance in crop type per state

- o Group bar chart to show poverty & crop production per state

Operations

- o Ability to filter states selecting what user wants displayed.
- o Hover options

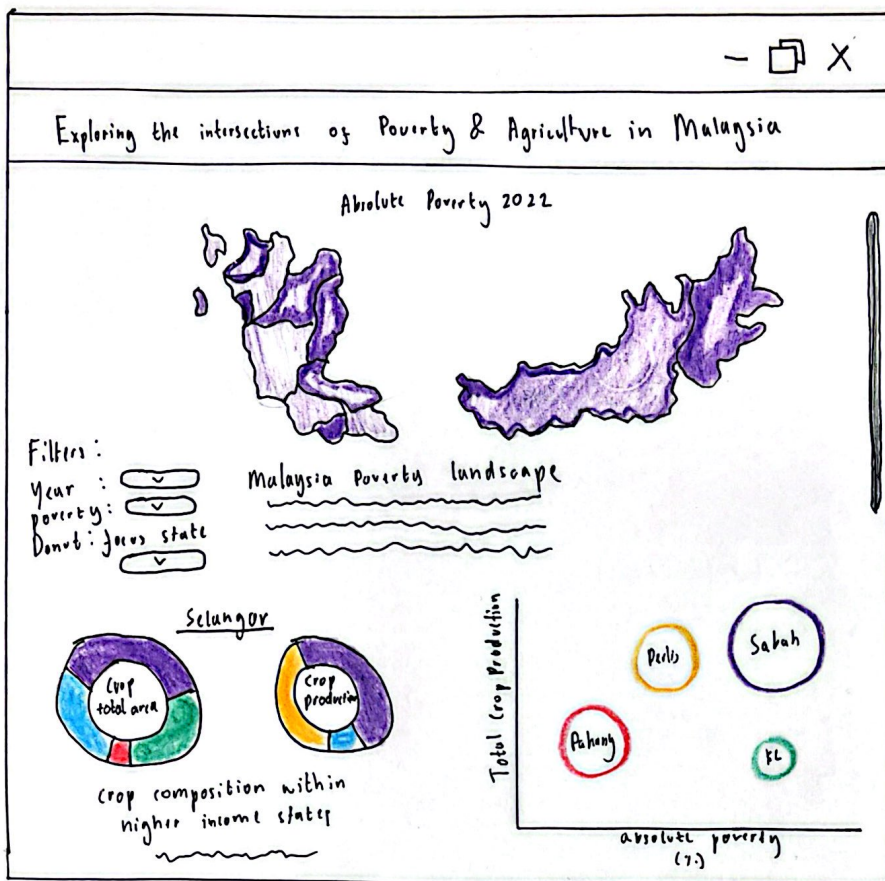
Focus



Details, Pros & Cons

Pros: Enables users to have clear visual comparison between poverty & crop output across states.

Cons: Can become cluttered or hard to interpret when too many categories or crops are shown.



focus

Jamie Ivanic Rajan.
 Title: Exploring the Intersection of Poverty & Agriculture in Malaysia
 Description: this sheet consolidates the full set of analysis across ~~time~~, ~~time~~ & their intersection: poverty, Agriculture

Operations

Operations/Components:

↳ Poverty:

Choropleth map: Visual of Malaysia
 time series: Change in poverty over the years.

Agriculture:

Donut chart: Composition of crop
 Scatter Plot: State crops by year.
 Combined
 Bubble chart: Show state similarity
 Grouped bar chart: Output & Poverty

Details

Exploring the Intersections of Poverty & Agriculture in Malaysia.

this visualisation explores how agricultural activity, crop composition & production efficiency relate to poverty across Malaysian states. Through six complementary idioms - spatial, temporal, & comparative.

this dashboard reveals patterns in agricultural distribution, economic disparity & possible relationships, between farming performance & poverty levels and whether more impoverished states focus on different crop types

Methods & Algorithms

- o Vega like to build visualisation
- o filtering
- o regression fitting
- o interactive viewing
- o R, group by(), summarise(), mutate()

Dependencies

- o R packages
- o Vega Libre
- o Data sources
- o Github pages

Estimates

- o learning: 4 hours
- o Idiom design: 6 hours
- o Dashboard layout: 3 hours
- o Refinement: 2 hours
- o total 15 hours

Requirements & Next steps.

- o data set must be consistent
- o Refine colour palette choices